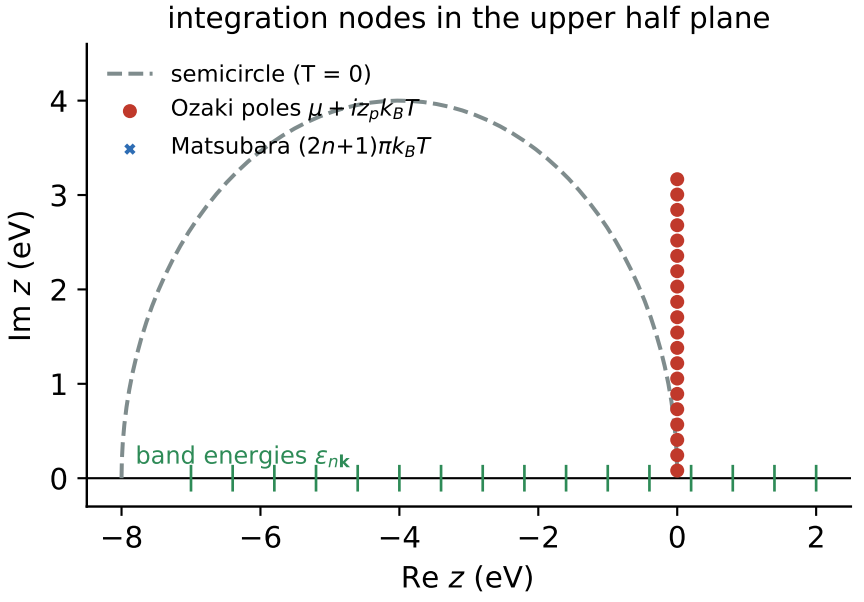
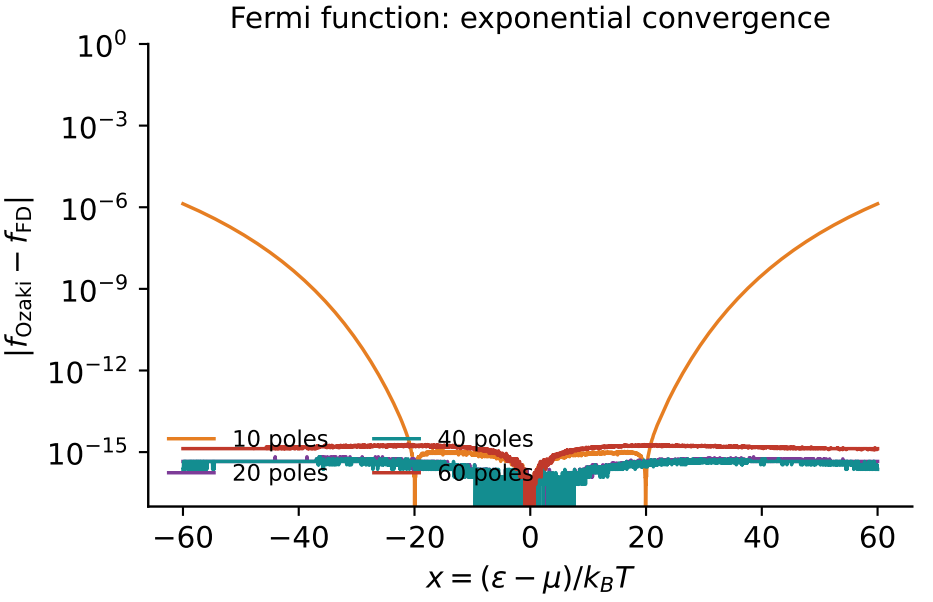

integration nodes in the upper half plane
semicircle (T = 0)
Ozaki poles $\mu + iz_p k_B T$
Matsubara $(2n+1)\pi k_B T$
band energies $\varepsilon_{n\mathbf{k}}$
Im z (eV)
Re z (eV)
4
3
2
1
0
−8
−6
−4
−2
0
2


Fermi function: exponential convergence
$|f_{\text{Ozaki}} - f_{\text{FD}}|$
10^{0}
10^{-3}
10^{-6}
10^{-9}
10^{-12}
10^{-15}
10 poles
20 poles
40 poles
60 poles
$x = (\varepsilon - \mu)/k_B T$
−60
−40
−20
0
20
40
60